The immediate advantage of the XGA projector is the small pixel size of the projected image, which lends it a smooth effect as opposed to the blocky SVGA detailing. Also, due to the increased number of pixels, the amount of aliasing (jagged edges) in geometric shapes is reduced.

All display systems except the CRT technology are plagued with the problem of "Native Resolution." This, essentially, is the default resolution at which the projector will give the best image quality. Even though it is possible to scale an SVGA projector to display at XGA resolution, this will severely affect image quality. Another important factor is that DLP projectors have an inherent edge over LCDs because they generally project a smoother picture and do not suffer from the grille artefacts seen in LCD projection systems. However, LCDs are known for their sharpness and more precise focusing—but this very attribute leads to more pixelation in movies and pictures.

Brightness

As a rule of thumb, it is advisable to get the brightest projector that your budget allows. Ambient light has a significant effect on the brightness. As far as totally dark rooms are concerned, you can get away with projectors with up to a 1000 lumen brightness rating. Projectors with a 2000 lumen rating are suitable for small conference rooms, where a little ambient light is tolerable. High-performance projectors with 3000 lumens are ideal for reasonably bright ambient lighting conditions. Such projectors are your best bet if you happen to travel around to different venues, where the audience, screen size and ambient lighting conditions vary. These will save you the embarrassment of poor visibility due to washed out images.

If, however, you want to cater to huge conference halls and auditoriums, you'll have to consider the ultra-bright range of projectors that starts at 3000 lumens and goes well over 12,000! You can even try using high-gain screens, which can substantially increase the existing brightness levels.

At the end of the day, you must realise that as brightness levels increase, the precision of black and greyscale levels decreases. A projector cannot reproduce black well in bright ambient lighting conditions. So if image accuracy is what you're looking for, we suggest a grey screen paired with a good DLP projector and zero ambient lighting.

When it comes to light efficiency, an LCD projection system traditionally has the upper hand over the DLPs. Also, the saturation levels of the colour rendition of LCDs are better than DLPs and hence their perceived brightness is much higher.

Contrast Ratio

The contrast ratio is the least important feature from the point of view of a business projector. It refers to the ratio of the dynamic range between the darkest black shade and the absolute brightest white. A projector with a high contrast ratio is capable of large number of colour gradations and grey scales. Business applications such as simple presentations, text, and graphs will not show any improvement on high-contrast projectors. But movies will. DLP projectors easily outclass LCDs when it comes to accurate black-level rendition.

When it comes to light efficiency, an LCD projection system traditionally has the upper hand over the DLPs

This is due to the reflective nature of the DLP, which allows the tiny micro-mirrors to completely divert the light away from the lens, to render perfect blacks with ease.

Weight

Business projectors need lightweight construction along with portability. DLP projection systems are generally lighter than their LCD counterparts, due to the compact digital nature of a single micromirror device at its heart, as opposed to the popular three-panel implementation found in most contemporary LCD projectors. However, portability doesn't come without its problems; the catch is the lower brightness associated with compact and lightweight projectors.

The Long And Short Of It

Different projectors ship with different kinds of lenses—short and long-throw lenses. We cannot comment about which is better because they cater to different projection needs, and are more about convenience than image quality.

A throw distance is the measure of separation required between the projector and the screen to project a certain area of image. Projectors with standard lenses need approximately two feet of distance between them and the screen to project an image of one square foot.

BenQ MP-610

Short-throw lenses, with their wider light dispersion characteristics, are able to project very large images over short distances. Long-throw projectors, on the other hand, are optimised with narrow dispersion characteristics to project images over large distances without losing much brightness. Most high-end projectors come with replaceable lenses that let you convert them to short or long throw on the fly. A short-throw projector is required in small conference rooms that don't have enough space to display large images. Long-throw projectors are useful in large auditoriums, where the projector needs to be placed well behind the audience.

Zoom Lenses

A zoom lens is a feature included in most projectors for convenience. A typical 1.2x zoom lens will give you a 20 per cent boost in image size, making it very handy in cramped spaces. However, this feature is pretty much useless if the zoom offered is digital, as opposed to the ideal optical zoom. Using digital zoom will only degrade the image quality.